Research Project Periodic Log Entry Form

01st -15th April

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
1 April	Conducted a group meeting to plan post-proposal activities and discussed system components including image processing and CNN-based classification.
3 April	Reviewed literature related to tea grading, CNN models, and image classification techniques based on proposal research.
4 April	Identified tea grade categories (Dust, Pekoe, BOP, FBOP) and studied their visual characteristics for dataset preparation.
8 April	Explored methods for image data collection under controlled conditions (lighting, background) as described in methodology
10 April	Designed initial data preprocessing plan including resizing, normalization, and augmentation techniques for improving model accuracy.
15 April	Researched suitable CNN models (MobileNet, ResNet, basic CNN) and discussed possible implementation approaches with the team.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:
- Code base checked : YES / NO

Signature : .....

Date : 20/04/2026

Research Project Periodic Log Entry Form

15th-30 April

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
16 April	Organized tea image dataset into categories (Dust, Pekoe, BOP, FBOP) and prepared data for training.
18 April	Applied image preprocessing techniques such as resizing, normalization, and augmentation to improve training quality.
19-20 April	Trained CNN model using Google Colab environment and tested initial classification performance on sample dataset.
22-24 April	Uploaded trained model file to Hugging Face and configured backend inference using Gradio.
26-28 April	Developed simple UI for image upload and connected frontend with deployed model API to generate predictions (as shown in Hugging Face Space).

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th May

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
01 May	Researched tea grading parameters and AI-based grading techniques.
03 May	Prepared tea grading datasets and image preprocessing pipeline.
06 May	Developed CNN-based tea grading classification model.
09 May	Integrated grading prediction APIs with backend services.
12 May	Demonstrated tea grading automation prototype during PP1 presentation.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th May

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
17 May	Improved tea grading model accuracy using additional training samples.
19 May	Enhanced grading classification logic and result consistency.
21 May	Improved grading result visualization and reporting features.
25 May	Integrated grading module with backend services and database.
29 May	Performed testing, debugging, and optimization of grading automation system.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2020

Research Project Periodic Log Entry Form

01st – 15th June

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
01 June	Expanded tea grading dataset with additional sample categories.
03 June	Improved grading model accuracy through model fine-tuning.
07 June	Enhanced grading result reporting and visualization features.
10 June	Integrated grading automation module with project database and APIs.
13 June	Performed system testing and corrected grading classification issues.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th June

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
16 June	Improved classification accuracy through hyperparameter tuning.
19 June	Enhanced image augmentation techniques for better model performance.
22 June	Evaluated model performance using validation datasets.
26 June	Improved prediction confidence and classification consistency.
29 June	Updated implementation documentation with recent improvements.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th July

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
02 July	Refined tea grade classification using additional validation images.
05 July	Improved prediction accuracy for multiple tea grades.
08 July	Enhanced grading result visualization on the web application.
11 July	Conducted functional testing of the grading workflow.
14 July	Resolved identified issues and optimized model performance.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 31st July

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
16 July	Integrated the grading model with backend APIs.
19 July	Improved processing speed for tea grade prediction.
22 July	Conducted end-to-end testing of grading and reporting features.
26 July	Optimized prediction workflow based on testing results.
30 July	Updated implementation documentation and testing reports.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th August

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
02 August	Refined the grading model using additional validation results.
05 August	Improved classification consistency across tea grades.
08 August	Enhanced grading dashboard and result presentation.
11 August	Conducted validation testing and resolved identified issues.
14 August	Prepared demonstration materials for Progress Presentation II (PP2).

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 31st August

Project ID : R26-IT-118

Student ID : IT22282668

Name : Pathirana I.M

Individual Component : AI-Based Tea Sorting and Grading Automation

Summary of RP work carried out during the reporting period

Date	RP work
16 August	Enhanced the AI-Based Tea Sorting and Grading module based on implementation feedback.
19 August	Improved module integration with the TeaGuard AI platform.
22 August	Conducted functional and integration testing of the grading module.
26 August	Updated technical documentation and presentation slides for PP2.
30 August	Finalized the Tea Sorting and Grading module for Progress Presentation II (PP2).

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026